

Is it Art, is it HCI? Exploring Tensions Between Practice and Research

Makayla Lewis
Kingston University
London, UK
M.M.Lewis@kingston.ac.uk

Josh Urban Davis
Dartmouth College
New Hampshire, United States
josh.u.davis.gr@dartmouth.edu

Eli Blevis
Indiana University
Bloomington, United States
eblevis@indiana.edu

Miriam Sturdee
University of St Andrews
St Andrews, Scotland
msturdee@gmail.com

Sarah Fdili alaoui
LRI
Paris-Saclay, France
saralaoui@lri.fr

Bill Gaver
Northumbria University
Newcastle, United Kingdom
w.gaver@northumbria.ac.uk

Mafalda Gamboa
Chalmers University of Technology
Gothenburg, Sweden
mafalda.gamboa@chalmers.se

Claire Elisabeth Ohlenschlager
Independent Artist
The Hague, Netherlands
c.e.ohlenschlager@gmail.com

Lian Loke
The University of Sydney
Sydney, Australia
lian.loke@sydney.edu.au

ABSTRACT

Art has the weight of our history as a people behind it, whilst in comparison, Human-Computer Interaction is relatively young. Artistic practice is a propeller of the innovations within HCI but works in this area often focus on the user study, the interaction, or the need for empirical evaluation. The plurality and tensions in art practice clash with this focus. Arts need not define the beholder as a 'user', but the addition of interaction and technology challenges its purpose. Aesthetics in isolation is not seen as impactful output, but as we investigate ways to bring art and HCI together, should we not focus on process and influence? This panel brings together a diverse group of artists, designers, practitioners, makers, and researchers, focusing on the challenges and variety of approaches in this space. The aim is to build dialogue to encourage a plurality of practices and creative responses in HCI.

CCS CONCEPTS

• **Human-centered computing** → **Human computer interaction (HCI)**.

KEYWORDS

Art, Sketching, Dance, Photography, Craft, Making, Sewing, Painting, Comics, Drawing

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1 INTRODUCTION

Human-computer interaction (HCI) has a rich history of interdisciplinary research borrowing from, and collaborating with, fields as diverse as psychology, material science, and biology. The arts are included in this breadth and diversity of subject matter, but there is a tension between what is defined as 'art', and what is defined as 'research' [1, 4, 5, 17]. It could be said that the line is drawn where there is, or is not, documentation of its reception or involvement of the 'user', but this divide is not so simple to establish. Art is often defined by process, materiality, and output, but equally, we might include design under its umbrella (especially research through design) as following a similar approach. How we define what 'art' is in HCI has a lot to do with how we acknowledge it within the research sphere e.g., [4, 10, 11, 15, 16, 18]. How we use art concerns our own competencies and connections. How we value the arts in HCI is a matter of personal approach and opinion. By bringing together those inside and out with this liminal arts space, we might discover how to make room for a new perspective. Art is a millennia-old practice that delights, engages, creates disagreement, forms inquiry, and can be shared, destroyed, or questioned. In this sense, it is valuable for HCI research, as it not only predates it but can support it in a multitude of different ways e.g., [11, 13, 19, 20, 22]. When we design and build a prototype, we wish to get feedback, much as an individual artist outside the space may also wish to hear what the audience thinks. Nearly all HCI research follows process, uses materials, iterates, and strives for novelty, creativity, and value. Many researchers are also artists [21], and the line between research and art is drawn in different ways. Some might have an art practice in isolation from their research [14], and although it can inform methods by means of flow and mindful practice, others may only use their art as part of their skill set for producing HCI research or teaching others [12]. Some researchers address both approaches [2, 3]. We now also have the confounding aspect of art that is created by artificial intelligence or robotics, imagery, or even theatre,

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or dance, e.g. [1, 23]. Is this third, non-human, actor part of the problem, or part of the solution? Does this devalue art for what we believe it is or has it found acceptance in HCI?

2 CONNECTION TO RECHINNECTING

We are ReCHInnecting and making new connections between the arts and HCI.

3 ORGANIZERS & MODERATORS

Miriam Sturdee (organizer and moderator) is a Lecturer at the University of St Andrews in Scotland, working on intersections of art, design, and computer science. She is a practicing artist and designer and has an MFA in Visual Communication. Her publications explore areas of futuring, sketching, and drawing, and most recently psychology. **Mafalda Gamboa** (organizer and moderator) is a PhD Student at Chalmers University of Technology. Her background in architecture seeded her deep interest in the tensions between Art and Science. Her work is dedicated to exploring methods on the verge between art and design using situated, first-person, and visual narratives. **Makayla Lewis** (organizer and moderator) is a Lecturer in Computer Science (User Experience Design) at Kingston University London specializing in design thinking for technology innovation. She is also a researcher with an interest in human factors in business, cybersecurity, smart money, and artificial intelligence, especially for people with disabilities. She is an accomplished visual recorder (sketchnotes), illustrator, and doodler whose practices are embedded in her teaching, research, and home life.

4 PANELISTS

Our panelists are drawn from a pool of international artists, researchers, and designers. Given the breadth of the areas that the arts in HCI cover, we propose to have an additional panelist to give the most comprehensive worldview and give every area a voice. We have grouped our panelists into three main areas: Image & Design; Photography, Documentary & Performance; and Human-Robot-Art Interaction & Future Media (see Figure 1). This panel aims to explore the history of arts-based approaches and outputs in HCI, but equally, its future and possible goals. The tensions and camaraderie between subjects, artists, designers, and researchers or practitioners will allow for lively discussion and inspiring ideas.

4.1 Image & Design

Bill Gaver is co-leading the Interaction Design Studio, which is a multidisciplinary team pursuing design-led research. Gaver states that he is not an artist but draws upon artistic sensibilities and approaches in his work and share more generally an approach to research-based practice. His work centers on making new computational products that explore people's values and engagements with the world as well as technological possibilities, and his processes include a variety of design-led methods for engaging with settings, exploring ideas, and understanding how people engage with his work. He has been engaged with the HCI community for over 20 years and have seen orientations to and acceptance of this style of work rise and fall repeatedly over this time. Much of his work has been embraced both within HCI and in the design community more widely. Recently, however, he observes that there has

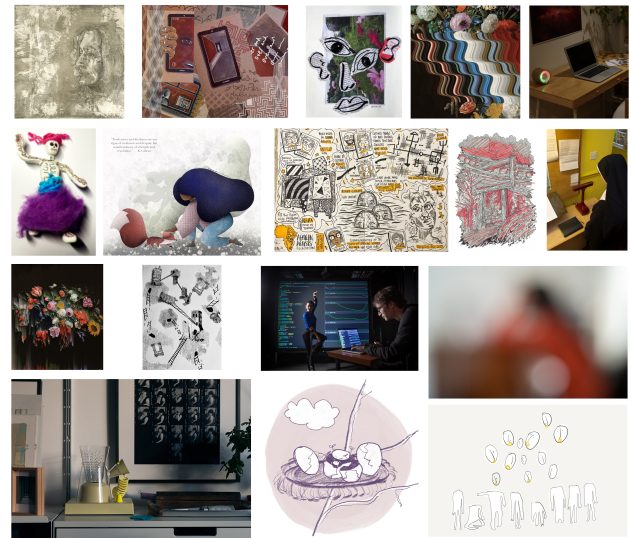


Figure 1: Authors art practice gallery (from left to right): Mopic, Aquatint Etching, Miriam Sturdee, 2010; A Dream, mixed media on paper, Miriam Sturdee, 2022; Observational mixed media collage from The Museo del Novecento, Milano Italy on Google Arts and Culture, mixed media on paper, Claire Elisabeth Ohlenschlager, 2020; Lightning fingering down the sky, oil and algorithmic manipulation, Josh Urban Davis, 2020; Light Touch, a simple communication device that allows people to trade lights over the Internet, ©Interaction Research Studio, Gaver et al. 2022 [9]; Observational mixed media collage from The Museo del Novecento, Milano Italy on Google Arts and Culture, clay on wire, string, felt and marker, Claire Elisabeth Ohlenschlager, 2020; "Tenderness and kindness are not part of despair", digital illustration, Makayla Lewis, 2021; Observational sketches from African Artists Foundation, Nigeria on Google Arts and Culture, Pen and Marker on Paper, Makayla Lewis, 2021; Overpass, Copic marker and Rotring Isograph on Bristol Board, Miriam Sturdee, 2011; The Prayer Companion, ©Interaction Research Studio, Gaver et al., 2010 [7]; A sudden rush of blood to the skin, oil and algorithmic manipulation, Josh Urban Davis, 2022; Hideouts No. 30: Colony Collapse, ink on paper, Josh Urban Davis, 2020; CO/DA, an improvisation practice involving dance, live coding, and movement sonification, photograph ©Christian Morel, LISN / CNRS Photothèque, Sarah Fdili Alaoui et al. [6]; Woman at a desk, photograph, Eli Blevis, 2017; Energy Babble is like an automated talk radio obsessed with energy issues, photograph by Alex Wilkie, ©Interaction Research Studio, Gaver et al. 2015 [8]; Illustration as a form of enquiry into futures and metaphors for human-robot interaction. Here, a drone is hatching its eggs, digital illustration, Mafalda Gamboa, 2021; Sketching as a practice to understand design knowledge, in this case, the separation of mind/body and how discourse about design often falls in this difficult negotiation, digital sketch, Mafalda Gamboa, 2022.

been an upsurge in 'scientific' demands for reporting which risks constraining and misrepresenting the value of creative practice.

Claire Elisabeth Ohlenschläger is a teacher and trainer of English at Hogeschool Rotterdam, The Netherlands. They are also an independent artist specializing in three-dimensional mixed media sculptures, digital collages, sketching, visual recording (sketch-notes), and black poetry. Claire work aims to help portray the main theme of a course or activity, for better understanding for viewers. They believe, "arts are a way to make people smile, to connect, to relax, to find inspiration and to inspire. Arts opens different perspectives, invites conversations, connects beyond borders and is an important way to convey ideas".

4.2 Photography, Documentary, & Performance

Sarah Fdili Alaoui is an associate professor at LISN-Université Paris Saclay in interaction design, human-computer Interaction, and dance. She is a choreographer, a dancer, and a Laban Movement Analyst. Sarah is interested in intersecting research in interaction design with dance-making and choreography. She has been involved in many art and science projects, collaborating with dancers, visual artists, computer scientists, and designers to create interactive dance performances, and interactive installations, as well as systems for supporting choreography and dance learning and documentation. **Eli Blevis** is a Professor of Informatics in Human-Computer Interaction Design (HCI/d) at the Luddy School of Informatics, Computing, and Engineering (Luddy) at Indiana University, Bloomington. His primary area of research, and the one for which he is best known, is sustainable interaction design. His research also engages visual thinking—especially photographic foundations of HCI, and design theory—especially transdisciplinary design.

4.3 Human-Art Robot Interaction & Future Media

Lian Loke is a researcher in the multi-disciplinary field of human-computer interaction studies and stages the interactivity of humans and machines through a choreographic and somaesthetic lens. Her work is characterized by the translation of embodied knowledge from dance and somatics into an aesthetics of interaction that radically integrates epistemologies of practice from the arts and sciences. Most recently she co-curated the SHE robots exhibition, the first of its kind in Australia to showcase the pioneering work of women in robotics across architecture, art, design, and technology. Dr. Lian Loke is an Associate Professor of Interaction Design in the School of Architecture, Design, and Planning, The University of Sydney. **Josh Urban Davis** is an American designer and early-career researcher from Texas whose research spans a wide spectrum of topics in human-computer interaction (HCI), with a current emphasis on generative design, novel creativity support tools, and inclusive technologies. Davis' creative projects explore the relationship between emerging technologies, social relationships, and identity. Recently, his work has received a grant from the Queer Arts Council and has been featured in the Queer Futures Journal by 3oC. His work has been exhibited at Grey Area Foundation for the Arts (San Francisco, California), Synapse Projects (Brooklyn, NY) Diverse Works (Houston, Texas), the Blaffer Art

Museum (Houston, Texas), Chandler Center for the Arts (Randolph, Vermont), and was featured in collaboration with the New School as part of the Venice Architecture Biennale 2021. He currently lives in Oakland, CA, with his cat, Nocturne, where he is pursuing a Ph.D. in computer sciences at Dartmouth. He is on the job market for 2023.

5 HYBRID PANEL FORMAT

5.1 Plan for Audience Interactivity & Engaging Audience Members

The panel will be hybrid format to allow online and remote attendees. Interaction with the audience is key for the panel. To encourage this we will collect prompts from both the remote and live audience using Mentimeter. A tool allows for the most popular questions to be ranked by large audiences, regardless of where they are, to connect with our theme. Moderators will encourage the audience to make use of their own artistic practice during the panel. The audience will be able to submit their artistic reactions to the conversations held by the panelists either using a hashtag on social media #ArtsAtCHI, or a submission to the panel Google Drive Gallery (link made available on the day), both options will be shared at the beginning of the panel. The moderators will project the collection of reactions toward the end of the panel. The number of moderators allows us to find a good balance between online and in-person, as well as make sure that everyone can contribute both with prompts for the audience but also artistic reactions.

5.2 Semi-Structured Questions for Panelists

Prior to the panel submission, each panelist was asked the following questions: What do arts mean to you? What do arts in HCI mean to you? What form of art do you do and why? What is your current process? How have you embedded your art practice into your work and or personal life? What impact does it have? What responses have you received regarding your art practice or process? The responses were vast and diverse ensuring the most appropriate panelists were brought together to discuss 'Is it Art, is it HCI?'. The answers also offer an opportunity for the panelists to engage with each other and for the moderators to support interactions with panelists on a deeper level. Furthermore, the following questions, if necessary, will be used to keep the panelist conversing with each other ensuring the panel remains engaging for the audience: Does involving HCI in the arts generate new approaches to accessibility? How can AI and human artists share practice? (How do you think text-to-image AI, text-to-music, and chat can impact our creative practice?) Are people in HCI hesitant to engage with you because of your artistic practice? Are there areas of HCI that you are hesitant to engage with? What could technology communities learn from creative practitioners and vice-versa? What motivations do you find to keep going when HCI rejects or questions artistic practice?

5.3 Panel Logistics

The panel will not require special seating, although A/V and a visualizer will be required to share artistic responses. There are no audience size limitations, all interested in art practice are welcome

both in-person and online. One student volunteer would be beneficial to support attendees to engage in Mentimeter and the gathering of artistic responses for the gallery.

5.4 Accessibility Statement

The panel will be designed and delivered to be attended by as many people as possible. All text and verbal utterances will be clear, simple to understand, and in English. All shared imagery will contain Alt Text and Captions with appropriate color contrast. Moderators and panelists will clearly verbalize all responses at an appropriate speed. Remote and in-person attendees will have access to closed captioning provided by the conference. Opportunities for support, questions, and comments will be present throughout the panel, provided by the moderators.

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